INCH-POUND

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FEDERAL SPECIFICATION

WEBBING, TEXTILE, (COTTON, ELASTIC)

This specification is approved for use by all Departments and Agencies of the Department of Defense (DoD).

1. SCOPE

1.1 Scope. This specification covers various types of textile, elastic cotton webbing.

1.2. Classification. The webbing will be available in the following types and classes, as specified.

1.2.1 Types and classes. Tables II and III provide differentiators for the types and classes listed below.

Type I	-	Woven	
Class 1	-	Unbleached, bleached or dyed	Water repellent, mildew resistant
Class 2	-	Unbleached, bleached or dyed	
Class 3	-	Unbleached, bleached or dyed	
Class 4	-	Unbleached, bleached or dyed	
Class 5	-	Unbleached, bleached or dyed	Mildew resistant
Class 6	-	Unbleached, bleached or dyed	
Class 7	-	Unbleached, bleached or dyed	
Class 8	-	Dyed	
Class 9	-	Bleached	
Class 10	-	Unbleached, bleached or dyed	
Class 11	-	Unbleached, bleached or dyed	
Class 12	-	Unbleached, bleached or dyed	

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Class 13	-	Unbleached	
Class 14	-	Unbleached, bleached or dyed	
Class 15	-	Unbleached	
Class 16	-	Unbleached	
Class 20	-	Unbleached, bleached or dyed	Water repellent, mildew resistant
Class 22	-	Unbleached, bleached or dyed	
Class 24	-	Unbleached, bleached or dyed	
Class 26	-	Unbleached, bleached or dyed	Water repellent, mildew resistant
Class 27	-	Unbleached	Mildew resistant
Class 28	-	Bleached or dyed	
Class 29	-	Bleached	
Type II	-	Braided	
Class 1	-	Unbleached, bleached or dyed	
Class 2	-	Unbleached, bleached or dyed	
Class 3	-	Unbleached, bleached or dyed	

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2. Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-3064 - Standard Practice Evaluation of Quality of Textile Materials

(Copies of this document are available online at https://quicksearch.dla.mil.)

2.2.2 Other government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract.

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ENVIRONMENTAL PROTECTION AGENCY (EPA)

Federal Insecticide, Fungicide and Rodenticide Act (FIFRA) (40 CFR Part 162) State Registration of Pesticide Products
Federal Insecticide, Fungicide and Rodenticide Act (FIFRA) as amended by the Food Quality Protection Act of 1996 and the Pesticide Registration Improvement Act of 2003

(Copies of these documents are available online at https://www.epa.gov/pesticides.)

FEDERAL TRADE COMMISSION

Rules and Regulations Under the Textile Fiber Products Identification Act

(Copies of this document are available online at https://www.ftc.gov.)

2.3 Non-government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

AATCC EP1 - Evaluation Procedure for Gray Scale for Color Change
AATCC EP2 - Evaluation Procedure for Gray Scale for Staining
AATCC EP8 - Evaluation Procedure for AATCC 9-Step Chromatic Transference Scale
AATCC EP9 - Evaluation Procedure for Visual Assessment of Color Difference of Textiles
AATCC TM8 - Test Method for Colorfastness to Crocking: Crockmeter
AATCC TM16.3 - Test Method for Colorfastness to Light: Xenon-Arc
AATCC TM20 - Test Method for Fiber Analysis: Qualitative
AATCC TM30 - Test Method for Antifungal Activity, Assessment on Textile Materials: Mildew and Rot Resistance of Textile Materials
AATCC TM61 - Test Method for Colorfastness to Laundering: Accelerated
AATCC TM70 - Test Method for Water Repellency: Tumble Jar Dynamic Absorption
AATCC TM169 - Test Method for Weather Resistance of Textiles: Xenon Lamp Exposure

(Copies of these documents are available online at https://www.aatcc.org.)

AMERICAN SOCIETY FOR QUALITY

ASQ/ANSI Z1.4 – Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at https://asq.org.)

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ASTM INTERNATIONAL

ASTM D76/D76M	- Standard Specification for Tensile Testing Machines for Textiles
ASTM D572	- Standard Test Method for Rubber – Deterioration by Heat and Oxygen
ASTM D573	- Standard Test Method for Rubber – Deterioration in an Air Oven
ASTM D1777	- Standard Test Method for Thickness of Textile Materials
ASTM D1907/D1907M	- Standard Test Method for Linear Density of Yarn (Yarn Number) by the Skein Method
ASTM D3775	- Standard Test Method for End (Warp) and Pick (Filling) Count of Woven Fabrics
ASTM D3776/D3776M	- Standard Test Methods for Mass per Unit Area (Weight) of Fabric

(Copies of these documents are available online at https://www.astm.org.)

INFORMA HEALTHCARE

Repeat Insult Patch Test – Modified Draize Procedure
Principles and Methods of Toxicology, A. Wallace Hayes (editor)

(Copies of this document are available online at https://www.crcpress.com)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations, unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1. First article. When specified (see 6.2) a sample of the webbing shall be subject to first article inspection in accordance with 4.2.

3.2 Standard sample. Unless otherwise indicated, the finished webbing shall match the standard sample for shade and appearance, and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.3 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.4 Materials.

3.4.1 Yarns. The webbing yarns shall be made from either carded or combed cotton yarn. The yarns shall be spun and twisted into ply yarns as specified in Tables II and III.

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3.4.2 Elastic strands. The elastic strands shall be made of heat-resistant, compounded natural rubber, synthetic rubber, or a mixture thereof. The strands shall be of the gauge specified in Tables II and III. Where a core is covered, it shall be wrapped with multiple ends of cotton yarns as specified in Tables II and III.

3.5 Color.

3.5.1 Color. The webbing shall be unbleached, bleached or dyed as specified in 6.2. When dying is specified, the cotton yarns shall be vat dyed to the color specified in the contract or purchase order (see 6.2) before weaving. Unless otherwise specified, when the webbing is water repellent or mildew resistant, the shade of the dyed webbing prior to the application of the finish shall match the standard sample. The shade of the water repellent and mildew resistant webbing after finishing shall be that resulting from the combination of the base shade (before finishing) and the color imparted by the finish.

3.5.1.1 Visual shade matching. The color and appearance of the webbing shall match the standard sample when tested as specified in 4.5.

3.5.1.2 Labile sulfur. Dyes and compounds containing elementary sulfur capable of oxidation to sulfuric acid shall not be used. The dyed webbing shall be "Free" of labile sulfur when tested as specified in 4.5.

3.5.2 Colorfastness. The finished webbing shall conform to the colorfastness requirements listed below in Table I when tested as specified in 4.5. Bleached and undyed webbing shall not be subjected to colorfastness testing.

TABLE I. Colorfastness requirements for dyed webbing.

Type and Class	Colorfastness to laundering (3 cycles) (color change and staining) (min.)	Colorfastness to accelerated weathering (40 AFU or 170 kJ/(m^2nm) @ 420 nm) 1/ (min.)	Colorfastness to light (40 AFU or 170 kJ/(m^2nm) @ 420 nm) (min.)
Dyed (except black) - Type I, Classes 1, 5, 6, 8, 10, 11, 13, 14, 15, 16, 20, 22, 24, 26, 27, 28 Type II, Class 3	3-4	3-4	-
Dyed (except black) - Type I, Classes 2, 3, 4, 7 and 12; Type II, Classes 1 and 2	3-4	-	3-4
Dyed (black) - All Types and all Classes (see 1.2.1)	2-3	2-3	-

1/ AFU - AATCC Fading Units

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3.6 Webbing weave construction.

3.6.1 Weave No. 1. Webbing used in Type I, Classes 2, 3, 4, 7, 11 12 and 26 shall consist of one ground warp, one (1) stuffer warp, one (1) elastic warp and one (1) filling. The ground warp shall weave in a two-up to two-down order, one (1) end per heddle, and one (1) pick, so as to produce a plain weave effect on the face and back of the webbing. The stuffer warp ends and elastic warp strands shall weave in such a manner as to be over all the back picks and under all face picks.

3.6.2 Weave No. 2. Webbing used in Type I, Classes 1, 5, 6, 13, 14, and 27 shall consist of one (1) ground warp (elastic strands wrapped with multiple cotton ends) and one (1) filling weaving plain.

3.6.3 Weave No. 3. Webbing used in Type I, Classes 9, 10 and 29 shall consist of one ground warp, one (1) sewing section warp, one (1) elastic warp and one (1) filling. The ground warp and sewing section warp shall weave in a two-up to two-down order. The elastic warp shall weave one-up and one-down, over all the back picks and under all face picks.

Class 9 webbing shall have four (4) sewing sections with each sewing section unit consisting of a minimum of six (6) ends.

Class 10 webbing shall have three (3) sewing sections with the outside sewing section a minimum of three (3) ends and the other two (2) sewing sections a minimum of four (4) ends.

Class 29 webbing shall have three (3) sewing sections with the outside sewing section containing a minimum of four (4) ends and the other two (2) sewing sections containing a minimum of six (6) ends each.

3.6.4 Weave No. 4. Webbing used in Type I, Classes 8, 15 and 16 shall consist of one ground warp and one (1) elastic warp, one (1) elastic warp (with strands wrapped) and one (1) filling. The ground warp shall weave in a two-up to two-down order and the elastic warp shall weave in such a manner as to be over all the back picks and under all face picks.

3.6.5 Weave No. 5. Webbing used in Type I, Classes 20, 22 and 24 shall consist of two (2) ground warps (one face and one back), one (1) elastic warp, one (1) elastic warp, one (1) stuffer warp, one (1) binder warp, and one (1) filling. The face and back warps of the webbing shall be crossed with picks so as to produce a plain weave effect on the face and on the back of the webbing. The stuffer warp and the elastic warp shall weave in such a manner as to be over all the back picks and under all face picks. A binder warp shall weave between the face pick and the back pick, over all the face picks and under the back picks.

3.6.6 Weave No. 6. Webbing used in Type I, Class 28 shall consist of one (1) ground warp, one (1) elastic warp and one (1) filling. The ground warp shall weave one-up and one-down. The elastic warp shall weave one-up and one-down in such a manner as to be over all the back picks and under all face picks.

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3.6.7 Weave No. 7. Webbing used in Type II, Classes 1, 2 and 3 shall be plain weave, two-over and two-under braided weave.

3.7 Physical requirements. The Type I and II webbings shall meet the physical requirements as specified in Table II and Table III respectively when tested as specified in 4.5.

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TABLE II. Physical requirements (finished), Type I - woven.

Type I	Weave No. 1/	Width (inches)	Thickness (inches) (min.)	Weight (ounces per linear yard) (min.)	Yarns, full width (min)				Yarns per inch (filling)	Yarn size (±2 counts/ply)		Elastic strand gauge
					Ground warp	Binder warp	Stuffer warp	Elastic strands		Warp	Filling	
Class 1	2	1/2 (±1/32)	0.040 – 0.060	0.25	18	-	-	7	50 3/	20/2	24/2	36 2/
Class 2	1	1/2 (±1/32)	0.042	0.30	21	-	6	8	100	24/2	24/2	30
Class 3	1	3/4 (±1/32)	0.042	0.43	30	-	9	11	100 12/	24/2	24/2	30
Class 4	1	7/8 (±1/32)	0.042	0.50	36	-	11	13	100	24/2	24/2	30
Class 5	2	7/8 (±1/32)	0.040 – 0.060	0.45	33	-	-	12	50 3/	20/2	24/2	36 2/
Class 6	2	1 (±1/16)	0.045 – 0.060	0.50	36	-	-	13	50	20/2	24/2	36 2/
Class 7	1	1 (±1/16)	0.042	0.57	42	-	13	15	100	24/2	24/2	30
Class 8	4	1-1/4 (±1/16)	0.050 – 0.065	0.54	26	-	-	14	52	24/2	20/2	50 2/
Class 9	3	1-1/4 (±1/16)	-	0.62	28	24 4/	-	14	60 5/	20/2	20/2	36 6/
Class 10	3	1-1/2 (±1/16)	0.050	0.65	26	11 4/	-	17	70 7/	16/2	12/1	44 2/
Class 11	1	1-1/2 (±1/16)	0.038	0.85	100	-	24	26	96 8/	24/2	24/2	30 9/
Class 12	1	1-1/2 (±1/16)	0.043	0.87	66	-	21	23	100	24/2	24/2	30
Class 13	2	1-3/4 (±1/16)	0.048	0.95	54	-	-	19	40	12/2	12/1	36 2/

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TABLE II. Physical requirements (finished), Type I - woven. - Continued

Type I	Weave 1/	Width (inches)	Thickness (inches) (min)	Weight (ounces per linear yard) (minimum)	Yarns, full width (min)				Yarns per inch (filling) (min.)	Yarn size (±2 counts/ply)		Elastic strand gauge
					Ground warp	Binder warp	Stuffer warp	Elastic strands		Warp	Filling	
Class 14	2	2 (±1/16)	0.048	1.25	63	-	-	22	40	12/2	12/1	36 2/
Class 15	4	3 (±1/8)	0.049	1.28	42	-	-	22	52	24/2	20/2	50 2/
Class 16	4	6 (±1/8)	0.049	2.60	82	-	-	42	52	24/2	20/2	50 2/
Class 20	5	1/2 (+1/16,-1/32)	0.090	0.55	53	7	6	28	60 11/	20/2	20/2	30
Class 22	5	1-1/2 (±1/16)	0.090	1.50	153	36	17	19	60	20/2 and 12/2	12/2	30
Class 24	5	1-1/2 (±1/16)	0.090	2.10	182	40	38	44	86	20/2 and 12/2	24/2	30
Class 26	1	5/8 (±1/32)	0.053	0.43	46	-	10	12	96	30/2	24/2	28
Class 27	2	5 (±1/8)	0.048	3.25	162	-	-	55	40	12/2	12/1	36 2/
Class 28	6	1-1/4 (±1/16)	-	0.50	35	-	-	12	40 10/	16/2	16/2	42
Class 29	3	1-1/4 (±1/16)	-	0.62	36	16 3/	-	14	60 11/	20/2	20/2	36 4/

1/ See 3.6 for weave number descriptions.
2/ Elastic thread shall be covered with a double spiral wrap with 1 to 4 ends in the bottom layer, and 1 to 4 ends spiraling in the opposite direction in the top layer.
3/ 50 yarns per inch of 24/2 yarn or 90 yarns per inch of 40/2 yarns with 2 yarns in a shed.
4/ Sewing section warp.
5/ 60 yarns per inch of 20/2 yarns or 120 yarns per inch of 20/1 yarn with 2 yarns in a shed.
6/ Core wrapped with minimum of 3 ends cotton on bottom cover and 1 end cotton in opposite direction as top cover.
7/ 70 yarns per inch of 12/1 yarn or 140 yarns per inch of nominal 20/1 yarn with 2 yarns in a shed.
8/ 96 yarns per inch of 24/2 yarn or 160 yarns per inch of 40/2 yarn with 2 yarns in a shed.
9/ Unwrapped elastic strands. The edges may be covered rubber.
10/ 40 yarns per inch of 16/2 yarn or 80 yarns per inch of 16/1 yarn with 2 yarns in a shed.
11/ 60 yarns per inch of 20/2 yarns or 120 yarns per inch of 40/2 yarn with 2 yarns in a shed.
12/ 100 yarns per inch of 24/2 or 136 yarns per inch of 40/2 with 24 yarns in a shed.

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TABLE III. Physical requirements (finished), Type II – braided.

Type II	Weave No. 1/	Width (inches)	Thickness (inches)	Weight (ounces per linear yard) (min.)	Yarns, full width (min)		Yarns per inch (filling) (min.)	Yarn size (±2 counts/ply)	Elastic strand gauge
					Elastic strands	Carriers			
Class 1	7	5/16 (±1/32)	0.035 – 0.050	0.16	8	17	68	20/2	42
Class 2	7	7/16 (±1/32)	0.035 – 0.050	0.20	12	25	68	20/2	42
Class 3	7	9/16 (±1/32)	0.035 – 0.050	0.24	16	33	68	20/2	42

1/ See 3.6 for weave number descriptions.

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3.8 Finish treatment. When specified (see 6.2), the webbing shall be treated with a water repellent, mildew resistant treatment or a combination thereof.

3.8.1 Water repellency (Type I, Classes 1, 20 and 26). Classes 1, 20 and 26 shall be treated with a water repellent finishing treatment and shall have a dynamic absorption of no greater than 40 percent when tested as specified in 4.5.

3.8.2 Mildew resistant treatment Type I Classes 1, 5, 20, 26 and 27). The webbing shall be treated with an organic mildew inhibitor that is EPA registered under the FIFRA and the observable fungi shall be no greater than "Microscopic Growth" (visible only under the microscope) after leaching when tested as specified in 4.5.

3.9 Tension.

3.9.1 Initial. All webbing shall meet the tension and elongation requirements specified in Table IV, when tested as specified in 4.5 using the carriage load and elongation specified for the applicable class in Table IV.

3.9.2 Tension after accelerated aging.

3.9.2.1 Tension after accelerated aging (except Type I, Class 5). All classes, except for Type I, Class 5 shall change no more than 20.0 percent from the initial tension after being subjected to accelerated aging testing using the oven method when tested as specified in 4.5.

3.9.2.2 Tension after accelerated aging (Type I, Class 5). Type I, Class 5 shall change no more than 20.0 percent from the initial tension after being subjected to accelerated aging testing using the oxygen bomb method when tested as specified in 4.5.

3.10 Permanent set.

3.10.1 Initial permanent set. The permanent set of all webbing shall not exceed 8.0 percent initially when tested as specified in 4.5.

3.10.2 Permanent set after accelerated aging.

3.10.2.1 Permanent set after accelerated aging (except Type I, Class 5). The permanent set for all classes, except for Type I, Class 5 after being subjected to accelerated aging testing using the oven method shall be no more than 20 percent when tested as specified in 4.5.

3.10.2.2 Permanent set after accelerated aging (Type I, Class 5). The permanent set for Type I, Class 5 after being subjected to accelerated aging testing using the oxygen bomb method shall be no more than 20 percent when tested as specified in 4.5.

3.10.3 Permanent set after low temperature (Type I, Class 5 only). The permanent set of Type I, Class 5 webbing at -25°F shall be no more than 35 percent when tested as specified in 4.5.

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3.11 Elongation of all the webbing at -40°F(except Type I, Class 5, 28 and 29). Elongation of all the webbing (except Type I, Class 5, 28 and 29) at -40°F shall be not less than 10 percent when tested as specified in 4.5. Type I, Class 28 and 29 are not subject to elongation at low temperature.

3.11.1 Elongation of Type I, Class 5 webbing at -25°F. The elongation of Type I, Class 5 webbing at -25°F shall be not less than 50 percent when tested as specified in 4.5.

3.12 Performance after subjecting to boiling water (Type I, Class 5 only). Type I, Class 5 webbing shall meet the stated requirements for the following characteristics after subjecting to boiling water as specified in 4.6.15 when tested as specified in 4.5.

a. Tension before and after aging (see 3.9.1 and 3.9.2.2)
b. Permanent set (see 3.10.1, 3.10.2.2, 3.10.3)
c. Elongation at -25°F (see 3.11.1)

TABLE IV. Tension and elongation load.

Type and Class	Carriage load, lbs.	Percent elongation	Tension, lbs.		Low temperature elongation load lbs.
			Minimum	Maximum	
Type I, Class 1	10	50	1.20	1.50	1.50
Type I, Class 2	15	50	1.30	1.60	1.80
Type I, Class 3	15	50	1.70	2.05	2.30
Type I, Class 4	15	50	2.05	2.45	2.60
Type I, Class 5	15	50	1.65	3.25	4.00
Type I, Class 6	15	50	2.30	2.75	2.60
Type I, Class 7	20	50	2.35	2.85	3.10
Type I, Class 8	10	50	0.75	1.10	1.00
Type I, Class 9	15	50	1.45	1.95	1.75
Type I, Class 10	15	50	1.30	1.60	1.60
Type I, Class 11	25	40	4.50	7.00	7.75
Type I, Class 12	25	50	3.60	4.45	4.70
Type I, Class 13	20	50	1.95	2.40	2.50
Type I, Class 14	20	50	2.35	2.85	2.75
Type I, Class 15	20	50	1.25	1.55	1.60
Type I, Class 16	10	50	1.05	1.35	3.00
Type I, Class 20	30	50	4.40	6.00	5.80
Type I, Class 22	40	40	6.10	7.50	15.25
Type I, Class 24	50	40	11.60	14.10	18.50
Type I, Class 26	15	50	2.10	2.50	2.30
Type I, Class 27	30	50	3.15	4.65	6.50
Type I, Class 28	15	50	1.15	1.75	-
Type I, Class 29	15	50	1.45	1.75	-

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TABLE IV. Tension and elongation load. - Continued.

Type and Class	Carriage load, lbs.	Percent elongation	Tension, lbs.		Low temperature elongation load lbs.
			Minimum	Maximum	
Type II, Class 1	10	50	0.70	1.00	0.90
Type II, Class 2	10	50	0.80	1.10	1.00
Type II, Class 3	15	50	1.00	1.50	1.10

3.13 Length and put-up.

3.13.1 Length and put-up all except Type I, Class 20. Unless otherwise specified in the contract or purchase order, the webbing shall be put up in rolls or double-headed spools; except for Type I, Class 20 webbing (see 3.13.1). Roll or spools shall be of total lengths as specified in Table V. Rolls or spools shall contain no more than three (3) pieces and the minimum length of any piece shall be one (1) yard. One end of each piece in the roll shall be marked with the number of pieces in the roll using a means that is removable. The use of adhesive tape, staples, sewing or other devices to fasten adjoining ends of pieces of the webbing shall be prohibited.

3.13.2 Length and put-up Type I, Class 20. Type I, Class 20 webbing shall be put-up on double headed spools only. Rolls or spools shall be of total lengths as specified in Table V. Rolls or spools shall contain no more than six (6) pieces and the minimum length of any piece shall be two (2) yards. One end of each piece in the roll shall be marked with the number of pieces in the roll using a means that is removable. The use of metal staples and other metallic devices to abutt pieces of the webbing on individual rolls shall be prohibited.

TABLE V. Minimum and maximum length of webbing on rolls or spools.

	35 to 37 yards	50 to 60 yards	95 to 105 yards
Type I, Class 1		X	
Type I, Class 2	X		
Type I, Class 3	X		
Type I, Class 4	X		
Type I, Class 5		X	
Type I, Class 6		X	
Type I, Class 7	X		
Type I, Class 8		X	
Type I, Class 9		X	
Type I, Class 10		X	
Type I, Class 11		X	
Type I, Class 12	X		
Type I, Class 13		X	
Type I, Class 14		X	
Type I, Class 15		X	

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TABLE V. Minimum and maximum length of webbing on rolls or spools. - Continued

	35 to 37 yards	50 to 60 yards	95 to 105 yards
Type I, Class 16		X	
Type I, Class 20			X
Type I, Class 22	X		
Type I, Class 24	X		
Type I, Class 26		X	
Type I, Class 27		X	
Type I, Class 28		X	
Type I, Class 29		X	
Type II, Class 1	X		
Type II, Class 2	X		
Type II, Class 3	X		

3.14 Fiber identification. Each webbing spool shall be labeled or ticketed for fiber content in accordance with the Rules and Regulations under the Textile Fiber Products Identification Act.

3.15 Toxicity. The finished webbing shall not present a health hazard and shall show compatibility with prolonged direct skin contact when tested as specified in 4.5. The use of chemicals recognized by the EPA as human carcinogens is prohibited.

3.16 Workmanship. The finished webbing shall conform to the quality of product established on this specification. The occurrence of defects as specified in 4.5 shall not exceed the quality acceptance levels as specified in the contract or purchase order.

4. VERIFICATION

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)
b. Conformance inspection (see 4.4)

4.2 First article inspection. A first article, submitted in accordance with 3.1, shall be inspected, examined for appearance and finished defects listed in 4.4 and tested for the characteristics as specified in 4.5.

4.3 Conformance inspection. Conformance inspection shall include the visual examination of 4.4 and the tests of 4.5. Sampling for inspection shall be performed in accordance with ASQ/ANSI Z1.4 and with acceptance quality limits (AQLs) as specified in the contract and/or order, except where otherwise indicated (see 6.2).

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4.3.1 Inspection conditions. Unless otherwise specified, excluded, amended, modified or qualified in this specification or applicable procurement documents (see 6.2), all inspections shall be performed in accordance with all the requirements of referenced documents.

4.4 Visual examination. Each spool or roll in the sample shall be examined yard by yard for defects in accordance with MIL-STD-3064, Type IV, except the defects specified in Table VI shall also be scored four (4) points for each yard in which they occur.

TABLE VI. Defects that score four (4) points.

Defect Description
Catch-cord missing (applicable to shuttleless looms only)
Catch-cord not interlacing with filling (bobbin looms only)
Uneven application of treatment (Classes 1a, 1b, 4 and 8)
Clearly noticeable crystallization of mildew inhibitor (Classes 1a, 1b, and 4)

4.4.1 Spool or roll identification and marking examination. During the yard-by-yard examination each spool or roll in the sample shall be examined for defects as specified in MIL-STD-3064.

4.4.2 Shade variation examination. During the yard-by-yard examination, each spool or roll in the sample shall be examined for shade variation as specified in MIL-STD-3064.

4.4.3 Length examination (individual and total yardage). During the yard-by-yard examination, each spool or roll in the sample shall be examined for length as specified in MIL-STD-3064.

4.5 End item testing. The webbing shall be tested for the characteristics listed in Table VII. The methods of testing specified, wherever applicable and as listed in Table VII shall be followed. All test reports shall contain the individual values utilized in expressing the final results. The sample unit shall be as follows: nine (9) linear yards untreated webbing; 10 linear yards of treated webbing and 15 yards of Type I, Class 5 treated webbing. The lot shall be unacceptable if one (1) or more sample units fail to meet any requirement specified. The sample size shall be in accordance with the following:

Lot size (yards)	Sample size (sample units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

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TABLE VII. End item tests.

Characteristic	Requirement Reference	Test method
Cotton yarns: Material Ply Count	 3.4.1 Table II and Table III Table II and Table III	 AATCC TM20 (see 6.3) Visual ASTM D1907/D1907M
Rubber yarns: Material Gage	 3.4.2 Tables II and III	 Visual count
Visual shade matching	3.5.1.1	4.6.1
Labile sulfur	3.5.1.2	4.6.2
Colorfastness: Light Laundering (3 cycles) Accelerated weathering	3.5.2	 AATCC TM16.3 1/ AATCC TM61, Option 2, 2/ AATCC TM169, Option 4 1/
Weave	3.6	Visual
Thickness (inches) Type I Type II	Table II and Table III	ASTM D1777
Weight, per linear yard: Type I Type II	 Table II Table III	ASTM D3776/D3776M, Opt C
Filling yarns per inch Type I Type II	 Table II Table III	ASTM D3775
Yarns full width Type I: Ground warp Binder warp Stuffer warp Elastic strands Type II: Elastic strands Carriers	 Table II Table III	 ASTM D3775 ASTM D3775
Width, (inches): Type I Type II	 Table II Table III	Calibrated measuring tape

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TABLE VII. End item tests. - Continued

Characteristic	Requirement Reference	Test method
Dynamic absorption, Type I, Class 1, 20 & 26	3.8.1	AATCC TM70
Mildew resistance, Type I, Class 1, 5, 20, 26 & 27 (observed fungus growth), 4/ After leaching 6/	3.8.2	AATCC TM30, Test II, 5/
Tension Initial After accelerated aging (all except Type I, Class 5) After accelerated aging (Type I, Class 5)	 3.9.1 3.9.2.1 3.9.2.2	 4.6.3 4.6.4 4.6.5
Permanent set: Initial After accelerated aging (all except Type I, Class 5) After accelerated aging, (Type I, Class 5) After at -25°F temperature, (Type I, Class 5 only)	 3.10.1 3.10.2.1 3.10.2.2 3.10.3	 4.6.7 4.6.8 4.6.9 4.6.10
Elongation at -40°F temperature (all except Type I, Classes 5, 28 and 29) Elongation at -25°F temperature (Type I, Class 5)	3.11 3.12	4.6.12 4.6.13
Boiling water immersion (Type I, Class5 only) Initial tension Tension after accelerated aging Initial permanent set Permanent set after accelerated aging Permanent set at -25°F Elongation at -25°F	 3.12 &3.9.1 3.12 & 3.9.2.2 3.10.1 & 3.12 3.12 & 3.10.1 3.12 & 3.10.2.2 3.12 & 3.11.1	 4.6.15 & 4.6.3 4.6.15 & 4.6.5 4.6.15 & 4.6.7 4.6.15 & 4.6.9 4.6.15 & 4.6.10 4.6.15 & 4.6.13
Toxicity	3.15	4.7

1/ Rated using the AATCC EP1, Gray Scale for Color Change
2/ Rated using the AATCC EP1, Gray Scale for Color Change and AATCC EP2, Gray Scale for Staining.
3/ Rated using the AATCC EP8, AATCC 9-Step Chromatic Transference Scale
4/ AATCC TM30, Test II will be used for First Article (see 6.3).
5/ Five (5) specimens shall be evaluated.
6/ Leaching shall be conducted in accordance with procedure in Appendix A1 of AATCC TM30, prior to testing.

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4.6 Methods of testing and inspection.

4.6.1 Visual shade matching evaluation. The color and appearance of the dyed webbing shall match the standard sample when viewed using the AATCC EP9, Evaluation Procedure for Visual Assessment of Color Difference of Textiles, Option C (see 6.4) with sources simulating artificial daylight D_{75} illuminant with a color temperature of 7500 (± 200) K illumination of 100 (± 20) foot candles, and shall be a good match to the standard sample under incandescent A illuminant with a color temperature of 2856 (± 200) K.

4.6.2 Presence of labile sulfur test. In the determination of presence of labile sulfur in textile materials with lead acetate, two (2) 1.50 (± 0.01) gram samples from each material submitted for evaluation shall be tested. Each of the two (2) samples shall be cut into very small pieces and placed into separate test tubes. The samples shall be submersed in a stannous chloride solution that contains 100 grams of stannous chloride crystals ACS in 100 milliliters of hydrochloric acid ACS (35 percent concentration) and 50 milliliters of distilled water. A filter paper wet out with a 5.0 percent lead acetate solution shall be placed over the top of the test tube. The lead acetate solution contains 5.0 grams of lead acetate CP reagent grade and enough distilled water to make up a 100 milliliter solution; if the solution is not clear add a few drops (one at a time) of glacial acetic acid until the solution is clear. The test tube containing the textile sample, stannous chloride and wet filter paper shall be heated over a low flame until the solution is boiling. The solution should not be heated for more than 15 seconds. A brown to black stain on the filter paper should be evaluated as follows:

Free - The filter paper shows no discoloration or staining of any kind.
Slight - The filter paper shows a light tan to light brown discoloration stain.
Moderate - The filter paper shows a dark brown discoloration stain.
Severe - The filter paper shows a black color stain.

The rating shall be recorded. The results shall be recorded as “pass” or “fail”.

4.6.3 Initial tension test. Five (5) specimens shall be prepared. Each specimen shall be 14 inches in length and the full width of the webbing, except for webbings more than 3inches in width, in which case the specimen shall be raveled of each side until 3 inches are centered in the webbing. Bench marks shall be exactly 10 inches apart on each specimen and sewn together on the marks to form a loop 10 inches in circumference. The loop specimen shall be placed over the pins attached to the clamps of the constant-rate-of-loading (force) machine in accordance with ASTM D76/D76M. The starting gage shall be five (5) inches and shall be measured from the farthest edges of each holding pin. See Table III for specified load and elongation. The loading curve shall be recorded on the fifth cycle and each cycle shall be 40 seconds to load and unload. The tension in pounds of the five (5) individual specimens shall be determined on the loading curve at the specified elongation. The results shall be averaged and then divided by two (2) and reported to the nearest 0.01 pounds and recorded as “A”

4.6.4 Tension after accelerated aging test (all except Type I, Class 5). Five (5) test specimens shall be prepared as specified in 4.6.3 and then subjected to accelerated aging in conforming with ASTM D573, except that the oven temperature shall be 300°F and the time exposure shall be

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2 hours. After aging, the specimens shall be relaxed under standard conditions until they reach equilibrium and then tested in accordance with 4.6.3 for tension. The tension results after accelerated aging shall be averaged and then divided by 2 and reported to the nearest 0.01 pounds and recorded as "B". The percent change in tension shall be calculated as specified in 4.6.6 and reported to nearest 0.1 percent.

4.6.5 Tension after accelerated aging test (Type I, Class 5 only). Five (5) test specimens shall be prepared as specified in 4.6.3 and then suspended for a continuous period of 96 hours in a closed chamber, and subjected to oxygen, having a pressure of 275 to 300 pounds per square inch and maintained at a temperature of 158°F (±2)°F, as specified in ASTM D572. At the end of a 96 hour period, the test specimens shall be removed from the chamber and relaxed under standards conditions until they reach equilibrium, and then tested in accordance with 4.6.3 for tension. The tension results after accelerated aging shall be averaged and then divided by 2 and reported to the nearest 0.01 pound and recorded as "B". The percent change in tension shall be calculated as specified in 4.6.6 and reported to the nearest 0.1 percent.

4.6.6 Calculation of percent change in tension. The percent change in tension after accelerated aging shall be calculated as follows:

$$\text{Percent change in tension after accelerated aging} = \frac{A-B}{A} \times 100$$

Where A = initial tension (lbs), see 4.6.3
B = tension after accelerated aging (lbs)

4.6.7 Initial permanent set test. Three (3) test specimens each measuring 14 inches shall be cut and marked so that a distance of 10 inches (measurement A) is between the gage marks. The specimens shall be then stretched 50 percent, held at the elongation for 24 hours under standard conditions, and then released and allowed to rest for 10 minutes. The distance between the gage marks of the three (3) specimens shall be re-measured, averaged, and recorded as measurement B. The percent permanent set shall be calculated as specified in 4.6.11 and reported to the nearest 0.1 percent.

4.6.8 Permanent set after aging test (all except Type I, Class 5). Three (3) test specimens each measuring 14 inches shall be cut and marked so that a distance of 10 inches (measurement A) is between the gage marks. The specimens shall be subjected to accelerated aging in accordance with ASTM D273, except that the oven temperature shall be 300 and the time exposure shall be 2 hours. After aging, the specimens shall be relaxed under standard conditions until they reach equilibrium and then tested for permanent set in accordance with 4.6.7.

4.6.9 Permanent set after aging test (Type I, Class 5 only). Three (3) test specimens each measuring 14 inches shall be cut and marked so that a distance of 10 inches (measurement A) is between the gage marks. The specimens shall be subjected to oxygen, having a pressure of 275 to 300 pounds per square inch and maintained at a temperature of 158°F (±2)°F, as specified in ASTM D572. At the end of a 96 hour period, the test specimens shall be removed from the

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chamber and relaxed under standards conditions until they reach equilibrium, and then tested in accordance with 4.6.7.

4.6.10 Permanent set at -25°F test (Type I, Class 5 only). Three (3) test specimens each measuring 7 inches shall be cut and 5 inches shall be marked off (measurement A) in the approximate center of each specimen. Stretch each specimen to approximately 75 percent and condition at -25 (±2)°F for a minimum of 72 hours, release, and after 30 (±2) seconds measure the distance between gage marks. The measurement of the three (3) specimens shall be averaged and recorded as measurement "B". The percent permanent set shall be calculated as specified in 4.6.11 and reported to the nearest 0.1 percent.

4.6.11 Calculation of percent permanent set. The percent permanent set shall be calculated as follows:

$$\text{Percent permanent set} = \frac{A-B}{A} \times 100$$

Where A = initial permanent set measurement and
B = permanent set measurement after aging

4.6.12 Elongation at -40°F test (all except Classes 5, 28 and 29). Three (3) test specimens each measuring 7 inches shall be cut and 5 inches shall be marked off (measurement A) in the approximate center of each specimen. The specimen shall then be exposed to a temperature of -40°F (±5)°F for 2 hours, after which at that temperature it shall be suspended from a clamp in such a manner as to allow a weight to be hung on the lower end. The elongation load shall conform to the requirements specified in Table IV. The weight shall be gradually lowered until the entire load is carried by the specimen. After 2 minutes, measure the distance between the gage marks. The measurements of the three specimens shall be averaged and recorded as measurement "B". The percent elongation shall be calculated as specified in 4.6.14 and reported to nearest 0.1percent.

4.6.13 Elongation at -25°F test (Type I, Class 5 only). Three (3) test specimens each measuring 7 inches shall be cut and 5 inches shall be marked off (measurement A) in the approximate center of each specimen. Stretch each specimen approximately 75 percent elongation, three times to break up the sizing. Condition the specimen (unstretched) at -25°F (±2)°F for a minimum of 72 hours. Following the conditioning and while maintaining the temperature at -25°F (±2)°F, the specimen shall be suspended vertically and a tension load of 4 pounds shall be attached. The measurements of the three specimens shall be averaged and recorded as measurement "B". The percent elongation shall be calculated as specified in 4.6.14 and reported to nearest 0.1percent.

4.6.14 Calculation of elongation at low temperature. The percent elongation shall be calculated as follows:

$$\text{Percent elongation (low temperature)} = \frac{A-B}{A} \times 100$$

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Where A = initial measurement and
B = measurement of elongation at low temperature

4.6.15 Boiling water pre-treatment (Type, Class 5 only). Approximately nine (9) yards of webbing loosely arranged shall be submerged in boiling water for 1 hour. Remove the webbing and air dry at room temperature. After webbing is dry it shall then be placed in standard conditions until equilibrium is reached prior to testing Specimens for test should be marked out on the sample prior to subjecting to boiling water.

4.7 Toxicity test. When required, (see 6.2) an acute dermal irritation study and a skin sensitization study shall be conducted. When the results of these studies indicate the material is not a sensitizer or irritant, a Repeat Insult Patch Test shall be performed in accordance with the Modified Draize Procedure (see 2.3). If toxicity requirement (see 3.15) can be demonstrated with historical use data, toxicity testing may not be required (see 6.2).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The webbing is for use in the manufacturing of military tentage, clothing and equipage items.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type and class of webbing required (see 1.2).
c. The specific issue of individual documents referenced (see 2.2).
d. When first article is required (see 3.1)
e. Conformance inspection acceptance quality limits (AQL) (see 4.3)
f. Inspection conditions (see 4.3.1)
g. When toxicity testing is required (see 3.15, 4.7)
h. Packaging (see 5.1)

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6.3 Certificate of compliance. The contracting activity may select to accept a certificate of compliance for any stated requirement after first article testing.

6.4 Visual shade matching. In 2019, Option A of AATCC Evaluation Procedure 9, Visual Assessment of Color Difference of Textiles was changed to Option C. NOTE: In case of confusion, the viewing geometry should be such that the specimen plane and illumination source are parallel to each other and aligned so that the light flux is incident at the center of the specimen plane, which is set at a 35 (± 5)°angle relative to the horizontal. The observer will view the specimens at a 90° angle, relative to the plane of the specimens.

6.5 Subject term (key word) listing.

Dynamic absorption test
Leaching test
Microscopic growth
Water Repellent

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - GL
Navy - NU
Air Force - 11

Preparing activity:
DLA-CT

Reviewing Activities:
Army - AR, MD
Navy - AS, MC

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